A counter is a device which stores the no of times a particular event or process has occured.

made from

no of ff decides max how many times a counter can count

If no of ff = n ∴ Max times a counter can count = 2^n

if FF = 2

then counter can count $2^2 = 4$ → mod 4

aka

0, 1, 2, 3

if FF = 3

then counter can count $2^3 = 8$ →

mod 8

0, 1, 2, 3, 4, 5, 6, 7

Types of counter:

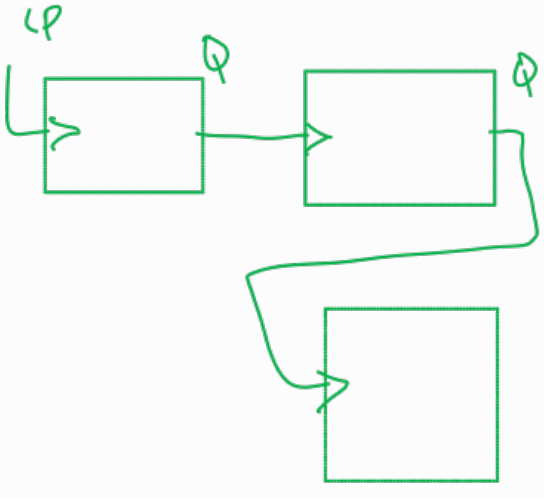
↙ ↘

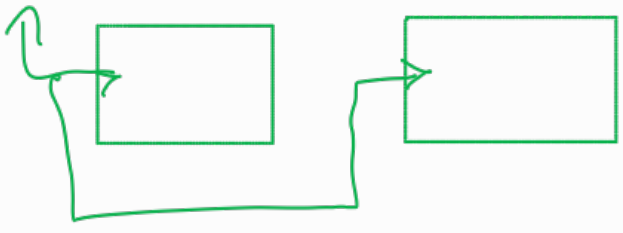
Asychronous counter aka Ripple counter

Not common clock pulse

Synchronous

common clock pulse given





(Very Slow)

(FAST) due to common clock pulse.

1) up counter

→ whose values increases. e.g Mod 8 up counter counting from (0 → 7)

2) Down counter

→ whose values decrease e.g Mod 8 Down counter.

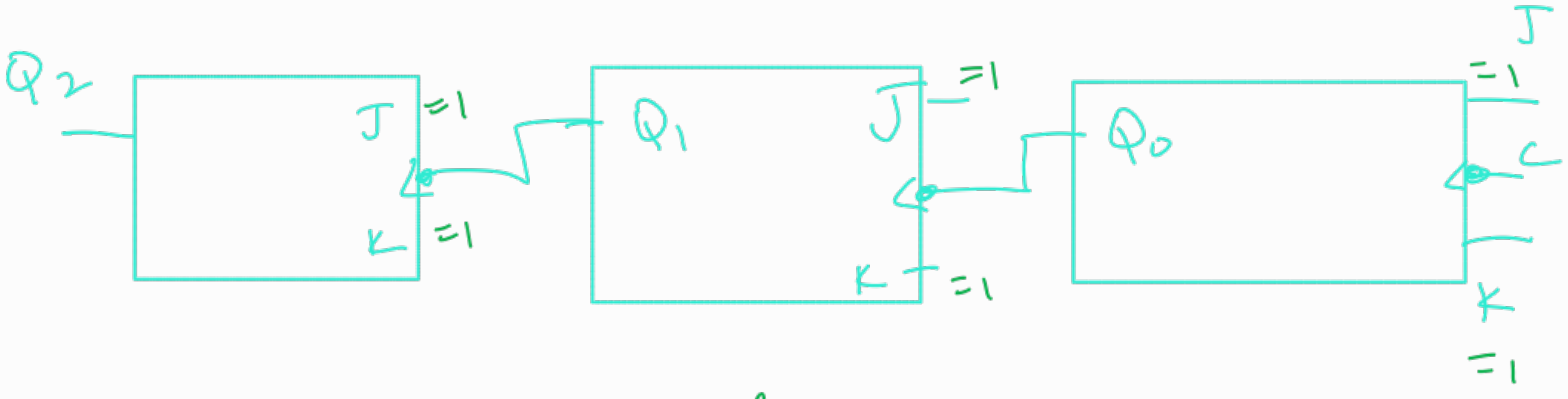
(7 → 0)

3) up down combination of Both (up and down)

---

3 Bit Asynchronous UP counter:

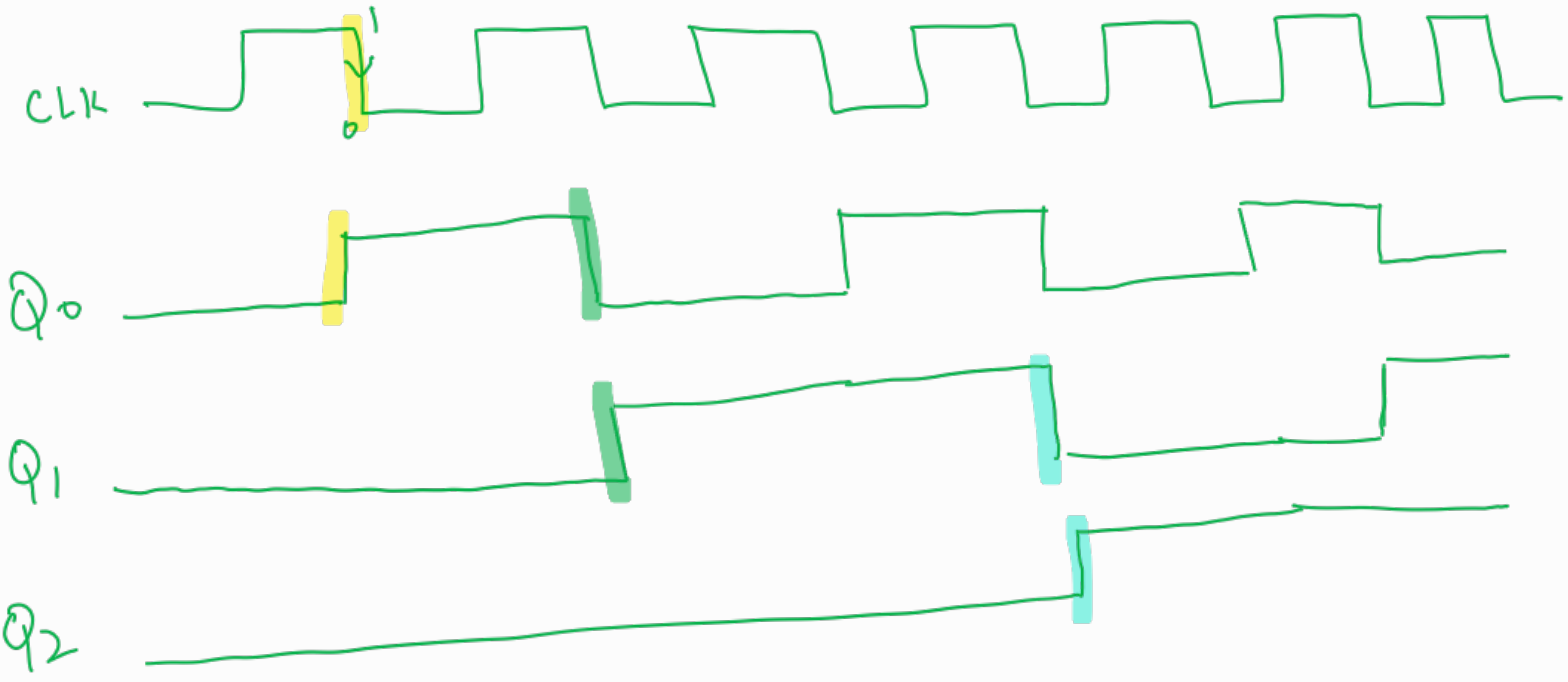
n no. of ff in which J=K=1 and is neg triggered then in total freq. $\left(\frac{\text{frequency}}{2^n}\right)$



we use toggle state for JK

J	K	Qt	Qt+1
1	1	0	1
1	1	1	0
1	1	0	1

works only on negative edge starting from LSB

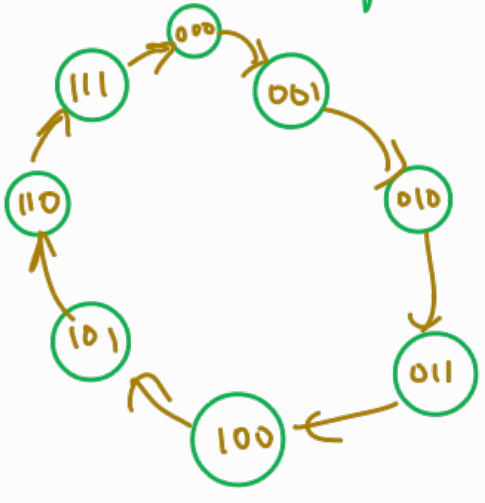


T.T

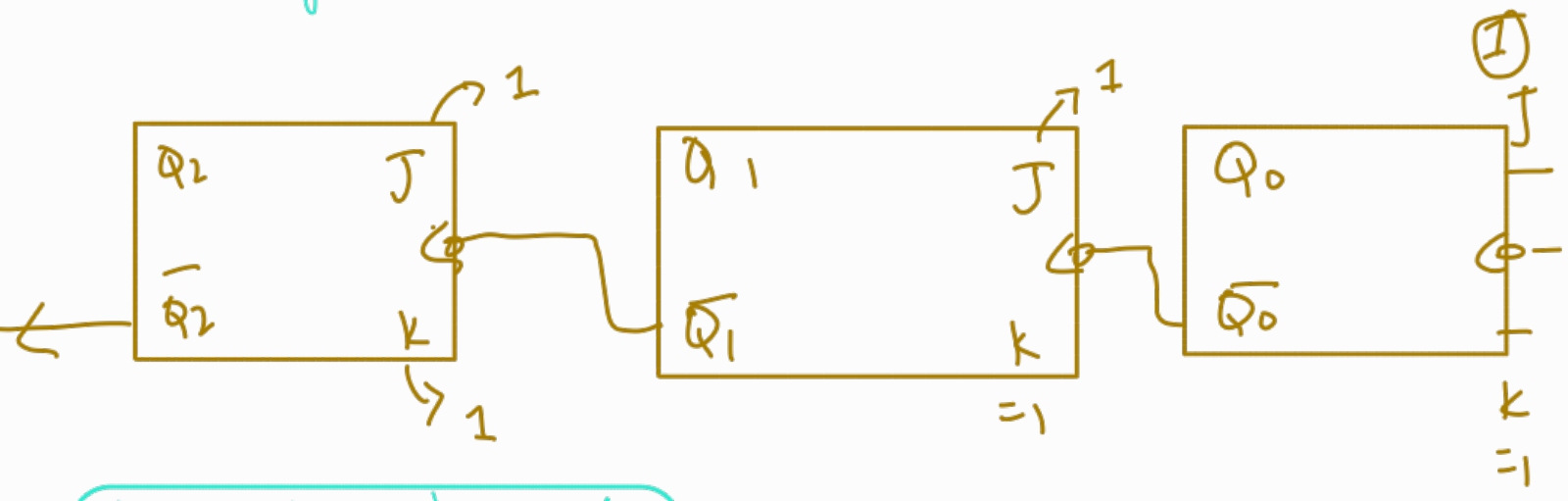
Q_2	Q_1	Q_0	
0	0	0	← (Intially)
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

with count from 0-7

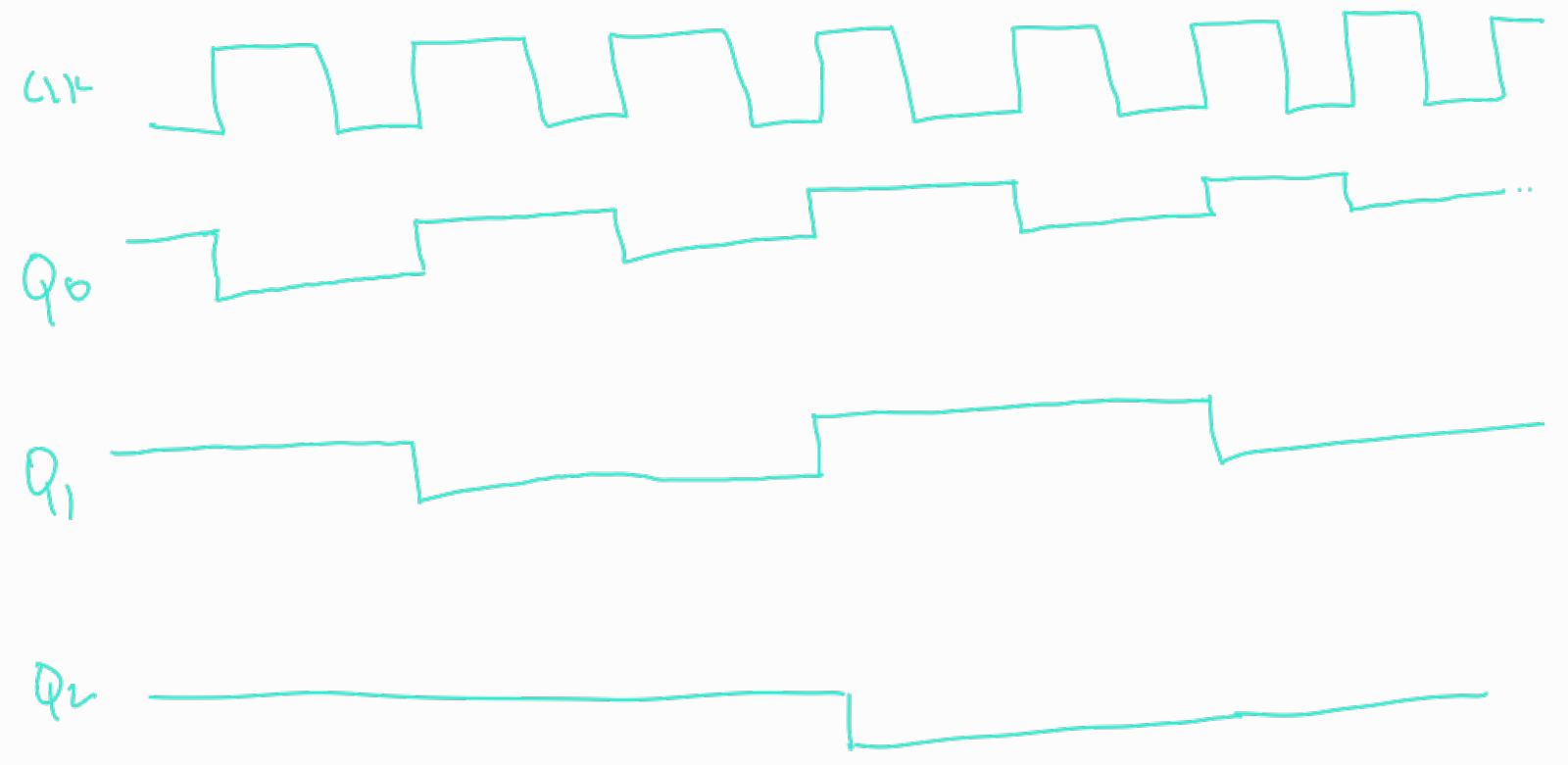
Stak diagram



3 Bit Asynchronous Down counter: (7-0)



neg edge triggered



(MSB) (LSB)

Q_2 Q_1 Q_0

✗ Draw Singles from T·T

Intial

1 1 1

Changes at + edge

1 1 0

1 0 1

Start T·T from (7) → (0)

1	0	0
0	1	1
0	1	0
0	0	1
0	0	0

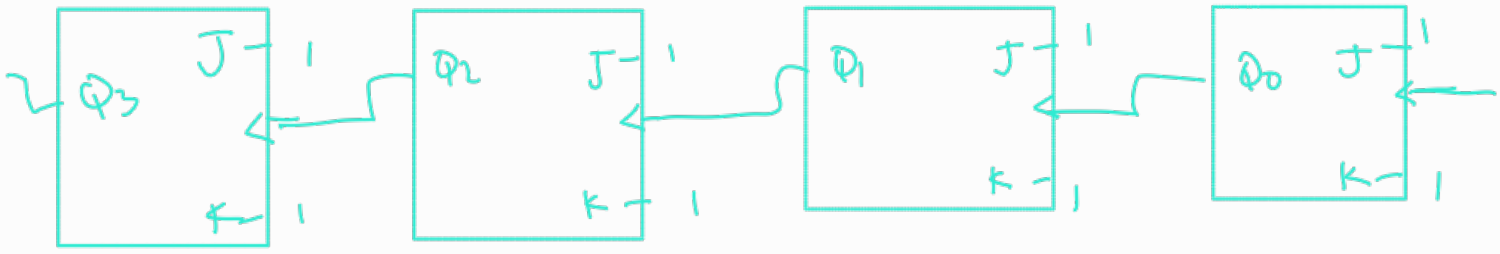
BCD Ripple counter (0-9) → Requires 4 bits

0000
0001
0010
0011
0100
0101
0110
0111
1000
1001

1 bit needs 1 FF for storing.

$2^4 = 2 \times 2 \times 2 \times 2 = 16$

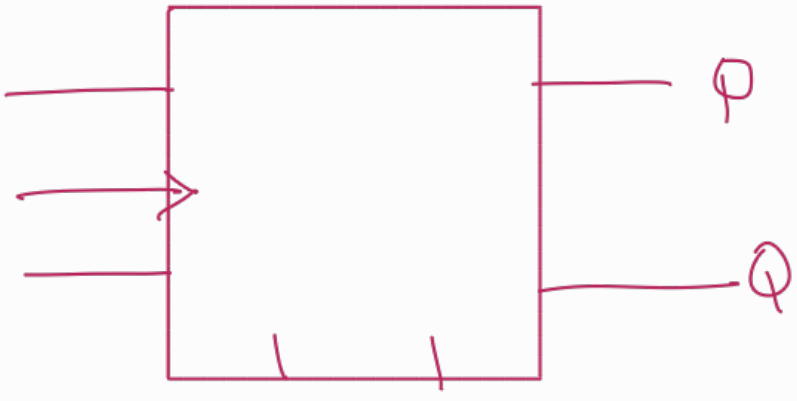
(0 - 15) counter.



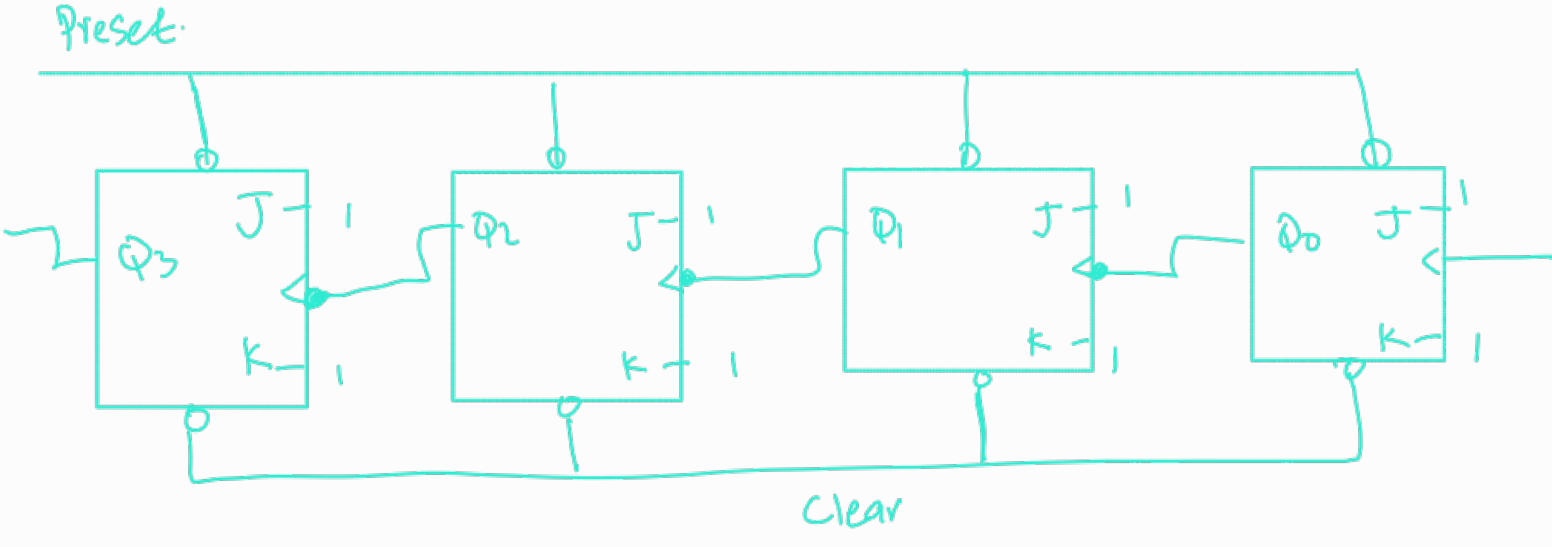
The T.T we will design will be of (0-15)
But as soon as >9 comes we will jump back to 0000. because its BCD ripple counter and it will count from 0-9 only.

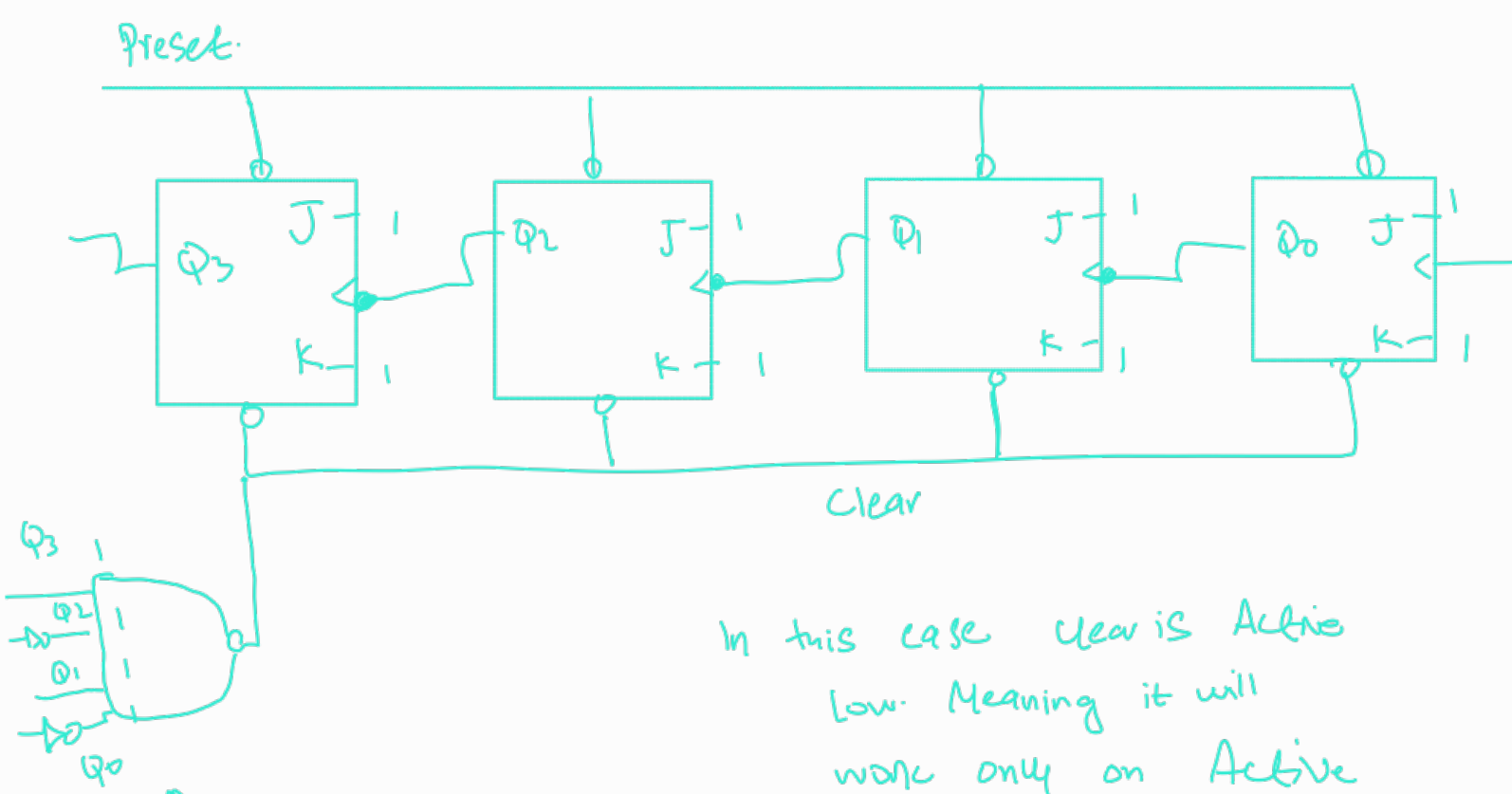
Clear | Preset

Clear pin → clears the present output.
Meaning Reset / Set to 0



Preset pin → Set the output to 1





In this case clear is Active Low. Meaning it will work only on Active Low

Q_3 Q_2 Q_1 Q_0 ← clear this.

1 0 1 0

$\overline{1 \cdot 1 \cdot 1 \cdot 1}$ → (0) → clear Actived due to Inverter.

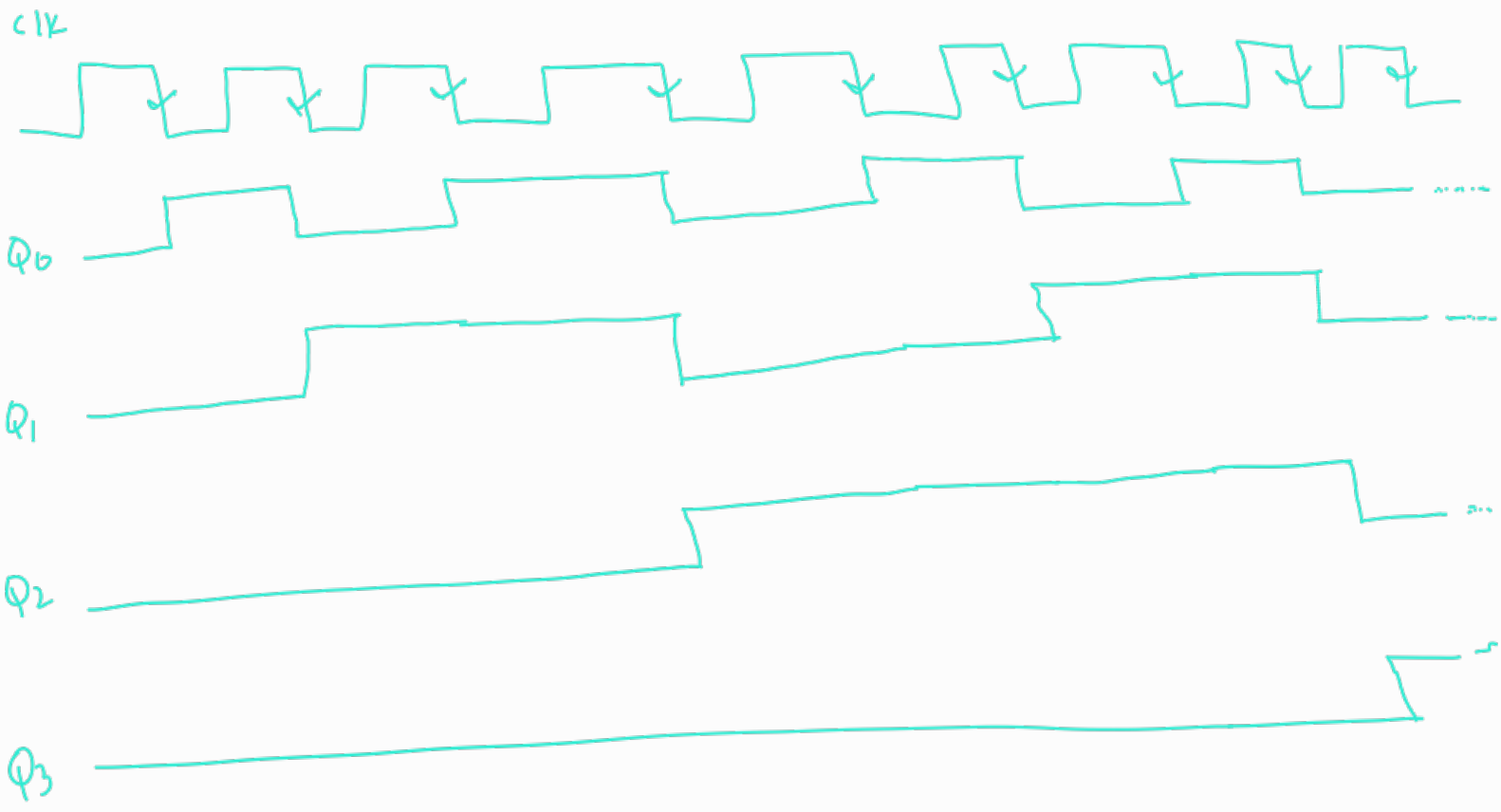
lets say

Q_3 Q_2 Q_1 Q_0

0 1 1 0

$\overline{0 \cdot 1 \cdot 1 \cdot 0}$ → (1)

Clear not Activited.



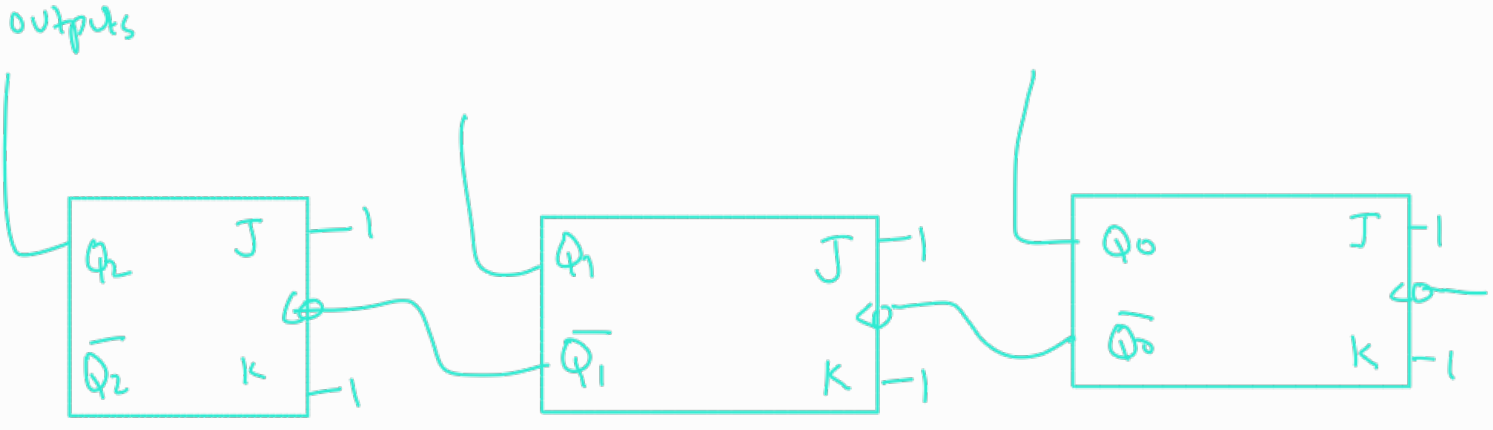
Q_3	Q_2	Q_1	Q_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0

1 0 0 1

last then it will repeat again

0 0 0 0

Binary UP/down counter → 3bit (0-7) (7-0)



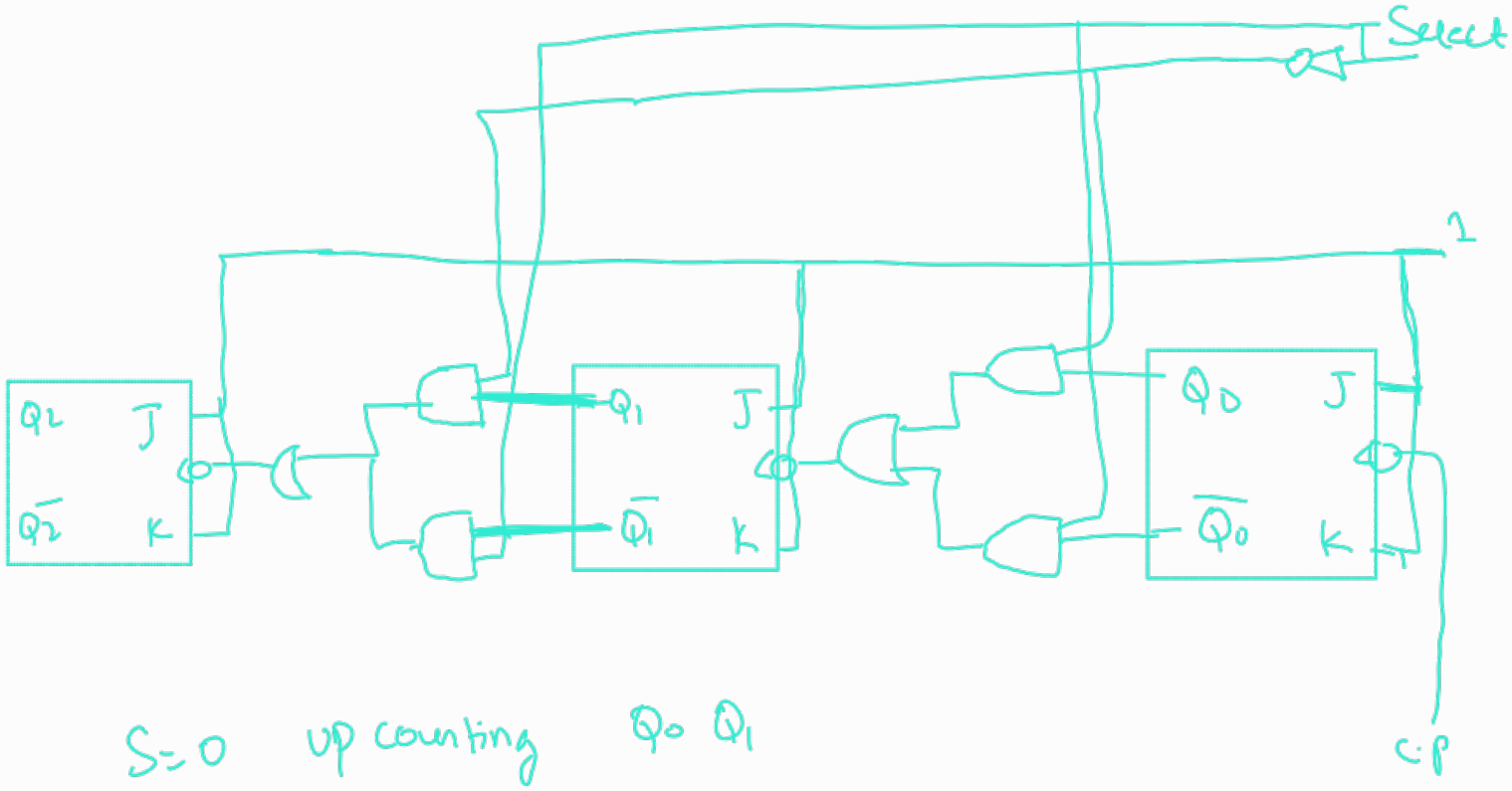
Down counters

Q_2	Q_1	Q_0	$\bar{Q}_2$	$\bar{Q}_1$	$\bar{Q}_0$
1	1	1	0	0	0

To count up we need to connect Q_n with clock pulse

To count down we need to connect $\bar{Q}_n$ with clock pulse

Now need to select any one of them. Mux



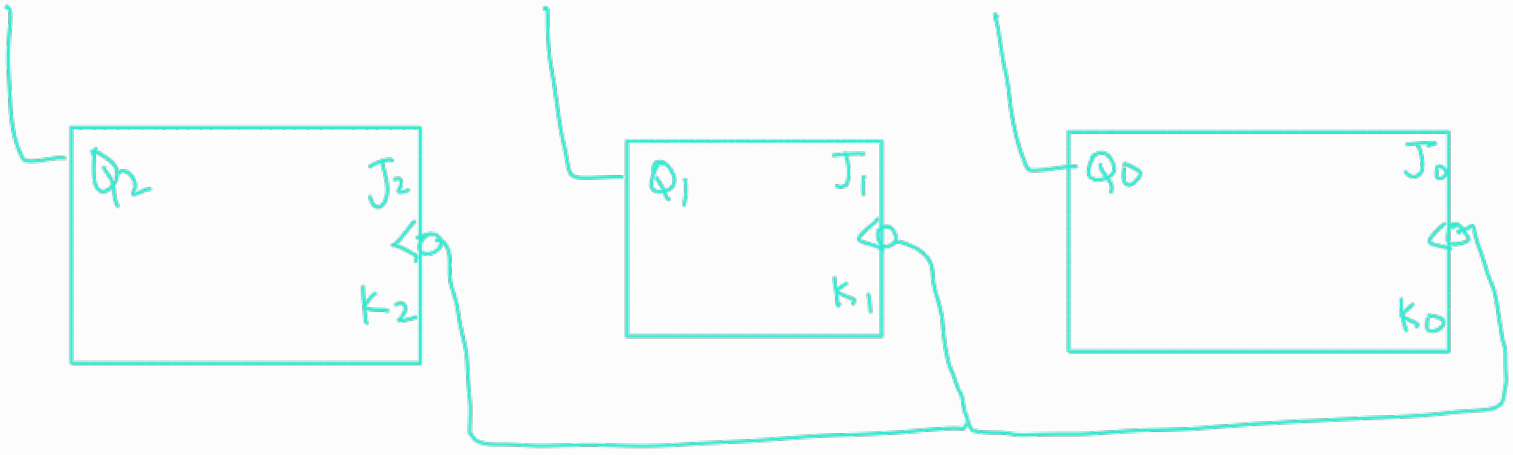
S = 0 up counting Q_0 Q_1

S = 1 down counting $\bar{Q}_0$ $\bar{Q}_1$

S	y
0	I_0
1	I_1

$$y = I_0 \bar{S} + I_1 S$$

3 bit Synchronous counter :: Up $2^3 = 8$ (0-7)

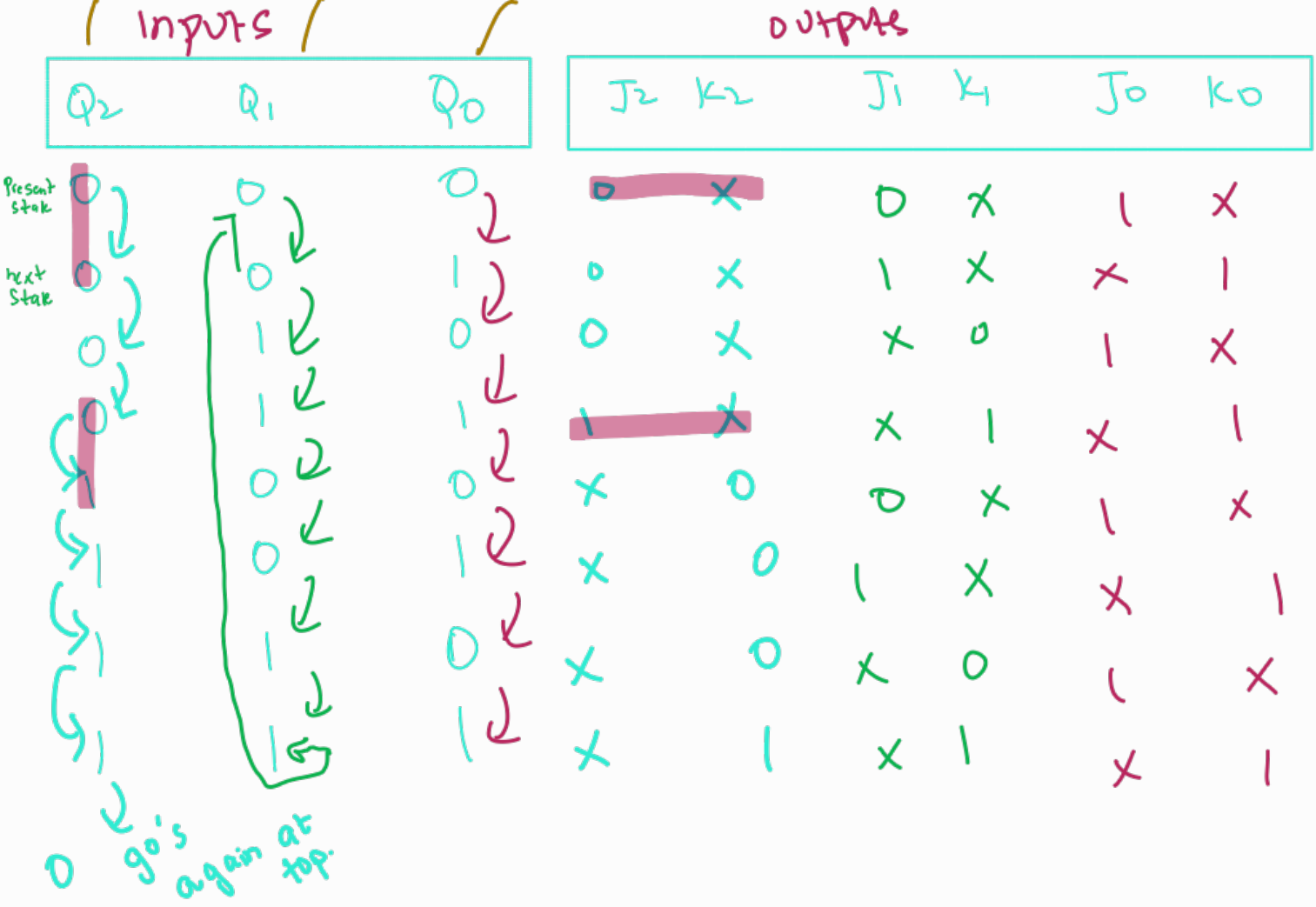


need excitation table for JK FF.

Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

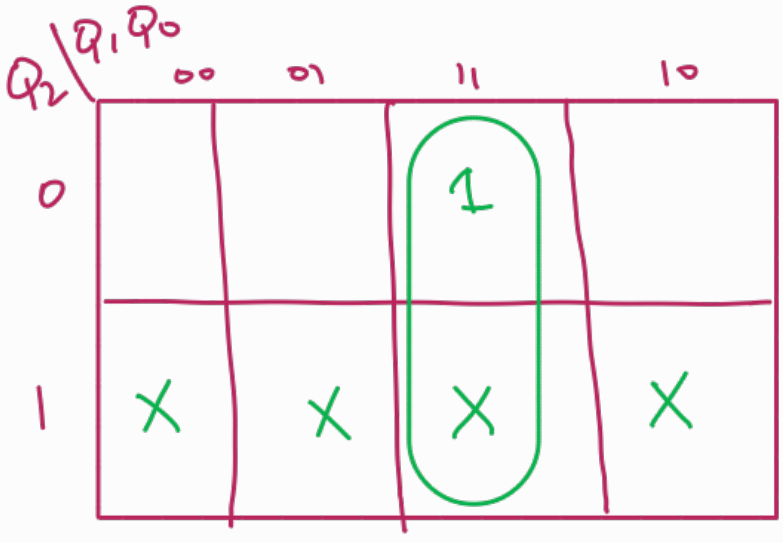
now we will fill the T.T by looking at

Starting from Q2 and fill J2 K2.

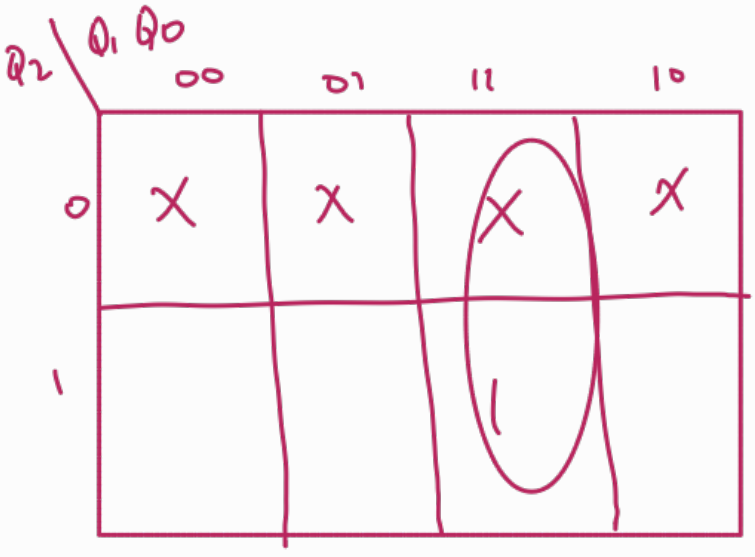


Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

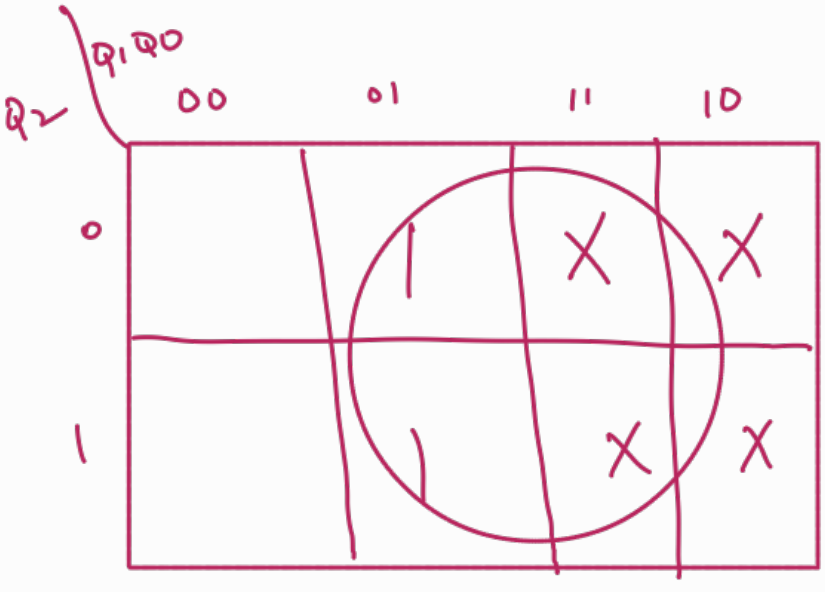
Making k-maps



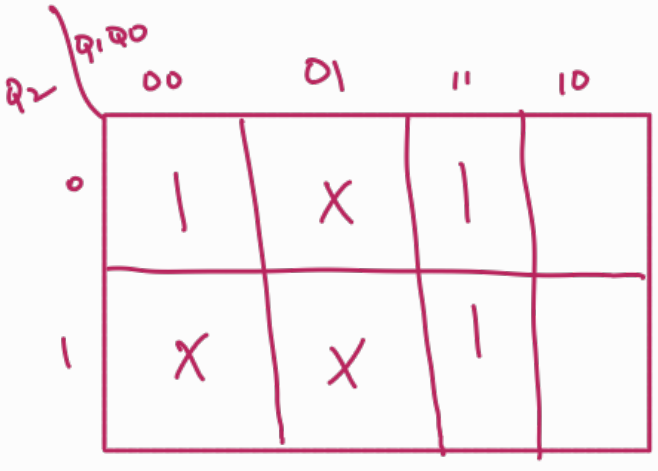
$J_2 = Q_1 Q_0$



$K_2 = Q_1 Q_0$



$J_1 = Q_0$

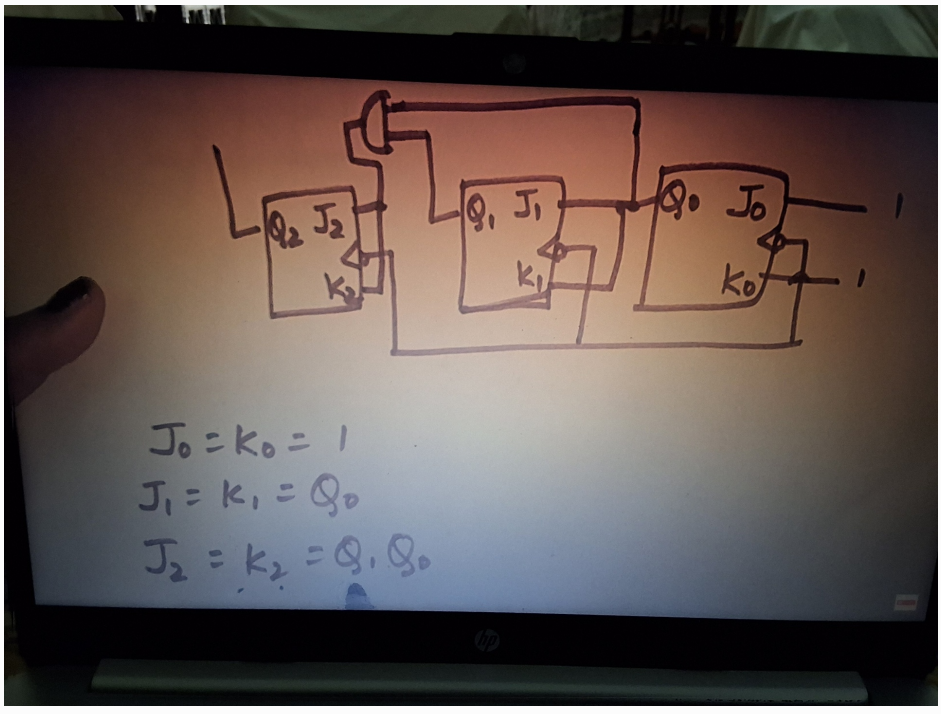


$K_1 = Q_0$

other equations →

$J_0 = 1$

$K_0 = 1$

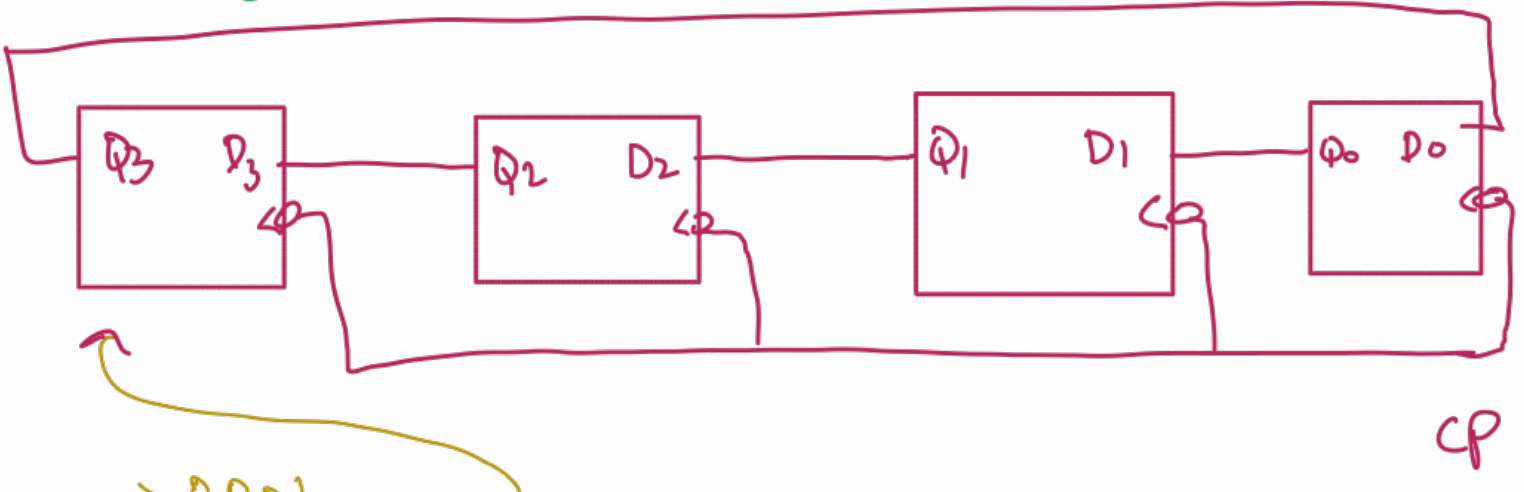


3 bit Synchronous counter

Ring counter: It is an application of shift Register

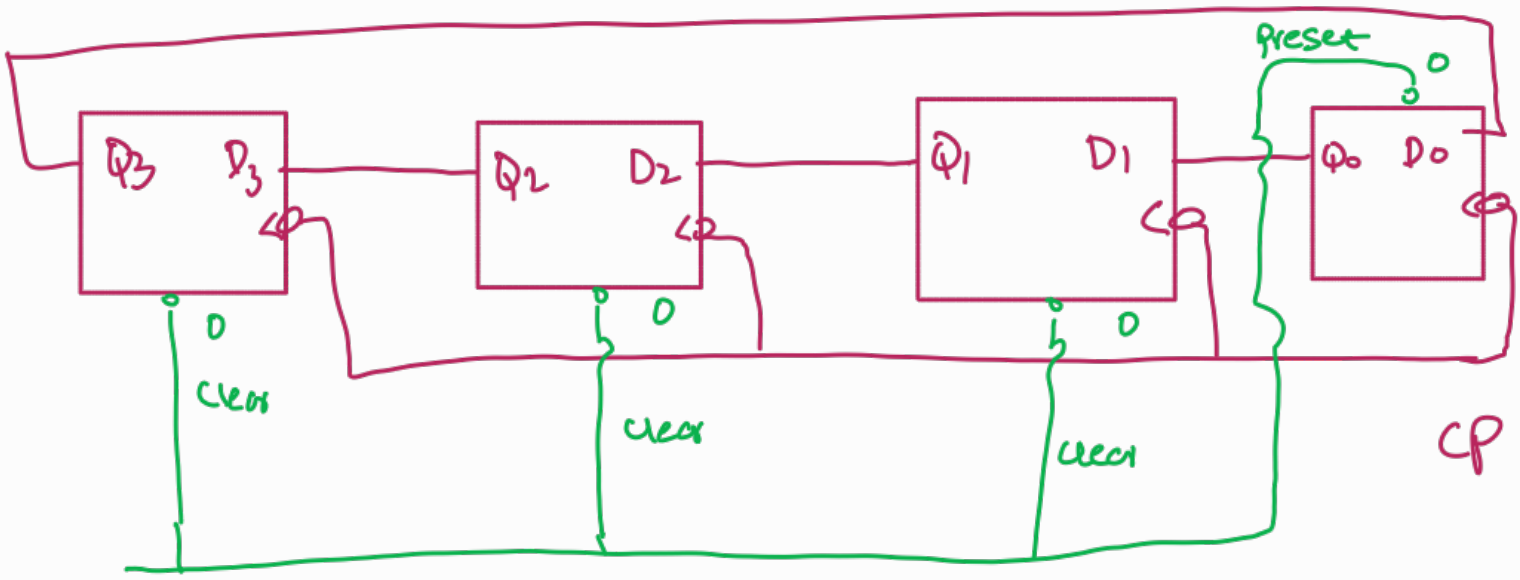
no. of States = no. of ff not (2^n)

Synchronous.



0001
0010
0100
1000
0001

State Recycles



0

when 0 is passed in ORI Q_3, Q_2, Q_1 = 0 and Q_0 = 1 due to preset

Initially

0 0 0 0

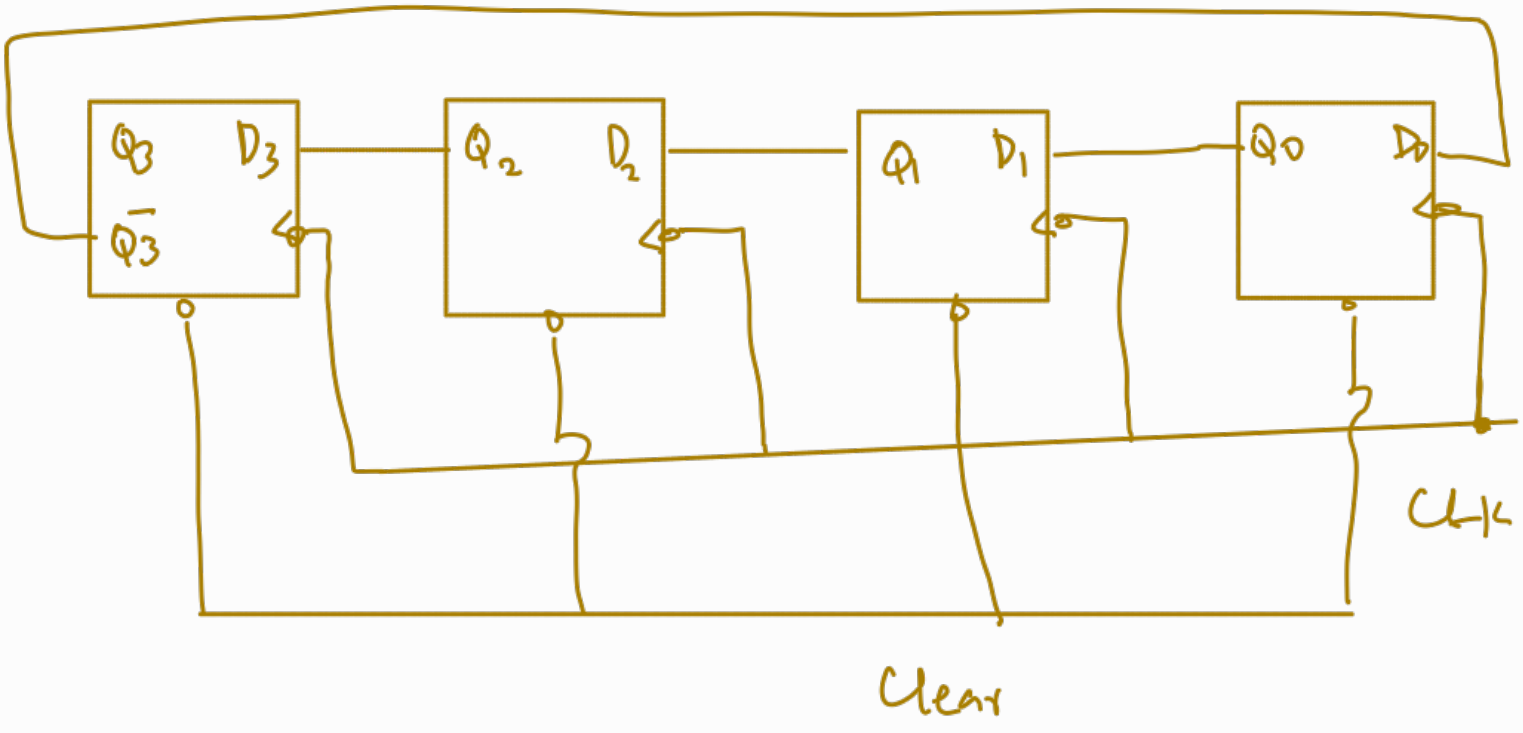
e.g Q_3 value goes to D_0

		Outputs				Inputs			
ORI	CP	Q_3	Q_2	Q_1	Q_0	D_3	D_2	D_1	D_0
0	X	0	0	0	1	0	0	1	0
1	↓	0	0	1	0	0	1	0	0
1	↓	0	1	0	0	1	0	0	0
1	↓	1	0	0	0	0	0	0	1
1	↓	0	0	0	1				
1	↓								

works on negative edge only.



Johnson counter: has greater number of states as compared to Ringcounter.



In J.C no need of Preset pin.

		inputs				outputs			
Clear	CP	D_3	D_2	D_1	D_0	Q_3	Q_2	Q_1	Q_0
0	X	X	X	X	X	0	0	0	0
1	↓	0	0	0	1	0	0	0	1
1	↓	0	0	1	1				
1	↓								
1	↓								

exactly copied

$Q_0 \rightarrow D_1$

$q_1 \rightarrow D_2$

$Q_2 \rightarrow D_3$

$\overline{Q_3} \rightarrow D_0$